Review of Sta4001: Stochastic Processes

Three important stochastic processes:

1) Discrete time Markov chains;

2) Poisson processes;

3) Continuous time Markov chains.

Major Models:

a. Gambler ruin problems;

b. Branching processes;

c. Random walks;

d. Queuing questions related to exponential distributions;

e. Birth-death processes;

f. Queuing models: $M/M/s$.

1. Probability distributions:

a. Discrete family: Bernoulli dist, binomial dist, geometric dist. Poisson dist, negative binomial dist.

b. Continuous family: uniform dist, gamma dist, beta dist, exponential dist, normal dist.

c. Multivariate dist:

• multinomial dist (rolling a k-sided die for n times, the distribution of the numbers of dot-1, dot-2, ..., dot-k).

• Dirichlet dist: joint pdf; it is the ratios of exponential r.v.'s and their sum.

d. Special properties of exponential distributions:

• The exponential distribution has the memoryless property, its remote family "discrete family" has a member with the same memoryless property: the geometric dist. We take advantage of the memoryless property of the exponential distributions in many occasions.

• The minimum of independent $X_i \sim Exp(\lambda_i), 1 \leq i \leq n$, is still exponentially distributed random variable with parameter $\lambda_1 + \cdots + \lambda_n$, and is independent of the ranks of X_i's.

• The sum of independent $Exp(\lambda_i), 1 \leq i \leq n$, in which all λ_i's are different, is a linear combination of their probability density functions.

• The exponential distribution is the building block of Poisson processes. It is also one of the two building blocks of a continuous time Markov chain: exponentially distributed holding times and a transition matrix.

• Conditional expectations and the formula of variance: variance = variance of conditional mean + mean of conditional variance.

2. Discrete time Markov chains

a. Basic definitions

• Transition matrix $P = (P_{ij})$ where $P_{ij} = P(X_{n+1} = j | X_n = i)$ for each $n \geq 0$.

• The k-step transition matrix (P_{ij}^k) of a Markov chain. Let $P_{ij}^k = P(X_k = j | X_0 = i)$. We know $(P_{ij}^k) = P^k$. As a result, P_{ij}^k is the (i, j)-entry of P^k.

• Chapman-Kolmogorov equation:

$$P_{ij}^{m+n} = \sum_k P_{ik}^m P_{kj}^n, \quad m, n \geq 1.$$

b. Classification of states of Markov chains.

• Communication between states: $i \longleftrightarrow j$; irreducibility, recurrence, transience, positive recurrence, periodicity, stationary distribution, ergodic property.

• Let $d_i = \gcd\{n \geq 1; P(X_n = i | X_0 = i) > 0\}$. With this notion, we say state i has period d_i. If $d_i = 1$, we say state i is aperiodic. Obviously, if $n = 1$ or there exists a prime number n such that $P(X_n = i | X_0 = i) > 0$, then state i is aperiodic.

• Let $f_i = P(X_n = i \text{ for some } n \geq 1 | X_0 = i)$. The state i is recurrent if $f_i = 1$; it is transient if $f_i < 1$. We have formula

$$\sum_{n=1}^{\infty} p_{ii}^n = \begin{cases} \frac{f_i}{1-f_i}, & \text{if } f_i < 1; \\ \infty, & \text{if } f_i = 1. \end{cases}$$

As a result a state i is recurrent iff $\sum_{n=1}^{\infty} p_{ii}^n = \infty$. A state i is transient iff $\sum_{n=1}^{\infty} p_{ii}^n < \infty$.

• Positive/null recurrences. For a recurrent state j, let $N_j = \min\{n \geq 1; X_n = j\}$ and $m_j = E(N_j | X_0 = j)$. We say state j is positive recurrent if $m_j < \infty$ and it is null recurrent if $m_j = \infty$. The 1-dim symmetric random walk is null recurrent.

• The states of 3-dimensional random walks are irreducible, and each state is transient.

• Ergodic chain: a Markov chain is irreducible, positive recurrent and aperiodic.

• A finite irreducible Markov chain is always positive recurrent.

• Class properties of Markov chains. Suppose $i \longleftrightarrow j$. We say a property is a class property if: i has the property implies that j has it. Recurrence, transience, positive recurrence, null recurrence have class properties.

c. Mean time spent in transient states.

A finite Markov chain has the set of transient states $T = \{1, 2, \cdots, t\}$. Define

$$P_T = \begin{pmatrix} P_{11} & P_{12} & \cdots & P_{1t} \\ \vdots & \vdots & \cdots & \vdots \\ P_{t1} & P_{t2} & \cdots & P_{tt} \end{pmatrix}.$$

Set $s_{ij} = E[\sum_{n=0}^{\infty} I(X_n = j)|X_0 = i]$, the mean time spent in state j after it starts at i. Define a $t \times t$ matrix $S = (s_{ij})$. Then $S = (I - P_T)^{-1}$.

d. Stationary distributions.

THEOREM 1 *(stationary distribution). Consider an irreducible, positive recurrent MC with transition matrix $P = (P_{ij})$. Write $\mathbf{1} = (1, 1, 1, \cdots)^T$ and $\pi = (\pi_1, \pi_2, \cdots)$. If the equation of π such that $\pi P = \pi$ and $\pi\mathbf{1} = 1$ has a solution, then the solution is unique. We call π the long-run proportion or stationary distribution. If there is no solution, the chain is either transient or null recurrent.*

• Fact: for a recurrent state j, define $N_j = \min\{n \geq 1; X_n = j\}$ and $m_j = E(N_j|X_0 = j)$. Then $\pi_j = \frac{1}{m_j}$ if j is positive recurrent and $\pi_j = 0$ if the chain is null recurrent.

THEOREM 2 *(law of large numbers/long-run proportion). Let X_n be an irreducible MC with stationary distribution $\pi = (\pi_0, \pi_1, \cdots)$. Let $r(x)$ be function with $E_\pi|r(X)| < \infty$. Then*

$$\frac{1}{N}\sum_{n=1}^{N} r(X_n) \to E_\pi r(X)$$

as $N \to \infty$. This implies that $\frac{1}{N}\sum_{n=1}^{N} \delta_{X_n}$ converges weakly to π.

THEOREM 3 *Let X_n be an irreducible and aperiodic MC. Then $\lim_{n\to\infty} P(X_n = j|X_0 = i) = \frac{1}{m_j}$ for any states i and j if*

(i) all states are transient or null recurrent in which $m_j = \infty$, or

(ii) all states are positive recurrent in which $\frac{1}{m_j} = \pi_j$. In this case the MC is ergodic.

e. Two important Markov chains.

• Random walks.

(1). For 1-dim random walk with $p = P(X_i = 1) = 1 - P(X_i = -1)$, where $\{X_i, i = 1, 2, \cdots\}$ are i.i.d. random variables. The random walk is $S_n = X_1 + \cdots + X_n, n \geq 1$. It is null recurrent if $p = 1/2$, and it is transient for $p \neq 1/2$.

(2). 2-dim symmetric random walk is null recurrent.

(3). A three or higher-dimensional symmetric random walk is transient.

• Gambler's ruin problem.

A gambler has i dollars. If he gambles, he will win \$1 with chance p and will lose \$1 with chance $q = 1 - p$. Then the chance he wins \$N before losing everything is

$$= \begin{cases} \frac{(\frac{q}{p})^i - 1}{(\frac{q}{p})^N - 1}, & \text{if } p \neq q; \\ \frac{i}{N}, & \text{if } p = q = \frac{1}{2}. \end{cases}$$

In the above formula, by swapping "p" and "q"; "i" and "$N - i$", we get the chance that the gambler loses everything before reach \$N.

3. Four Poisson Thinning Theorems

THEOREM 4 *(Poisson thinning without time). Suppose that the number of people who visit a yoga studio each day follows a Poisson distribution with mean λ. Suppose further that each person who visits is, independently, female with probability p or male with probability $1 - p$. Find the joint probability that exactly n women and m men visit the academy today.*

THEOREM 5 *(Poisson thinning for homogeneous Poisson processes). Consider a Poisson process $\{N(t), t \geq 0\}$ having rate λ, and suppose that each time an event occurs it is classified as either a type I or a type II event. Suppose further that each event is classified as a type I event with probability p or a type II event with probability $1 - p$, independently of all other events. Let $N_1(t)$ and $N_2(t)$ denote respectively the number of type I and type II events occurring in $[0, t]$. Then $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are two independent Poisson processes having respective rates λp and $\lambda(1 - p)$. The result can be generalized so that there are k types and each event is independently classified as type i with probability p_i for $1 \leq i \leq k$.*

THEOREM 6 *(Poisson thinning with time dependence). If the classification probability is time-dependent, and the event arriving at time t will be classified as type i with probability $P_i(s)$. Then, $N_i(t), i = 1, \ldots, k$, are independent Poisson random variables having means*

$$E\left[N_i(t)\right] = \lambda \int_0^t P_i(s) ds.$$

THEOREM 7 *(Poisson thinning with non-homogeneous Poisson process and time dependence). Let $N(t)$ be a Poisson process with intensity function $\lambda(s)$. If the classification probability is time-dependent, and the event arriving at time t will be classified as type i*

with probability $P_i(s)$. Then, $N_i(t), i = 1, \ldots, k$, are independent Poisson random variables having means

$$E\left[N_i(t)\right] = \int_0^t \lambda(s) P_i(s) ds.$$

4. Poisson processes.

A (homogeneous) Poisson process $X(t)$ with parameter λ is a special continuous Markov chain taking values $0, 1, 2, \cdots$. It has properties as follows.

• For each $t > 0$, $X(t) \sim Poi(\lambda t)$.

• Let $S_1, S_2, \cdots$ be the arrival times of the first event, second event, etc. Then the inter-arrival times $T_1 = S_1, T_2 = S_2 - S_1, T_3 = S_3 - S_2, \cdots$ are i.i.d. $Exp(\lambda)$-distributed random variables.

• The Poisson process $X(t)$ with parameter λ has the following three equivalent characterizations:

a. $X(t)$ has stationary independent increments and $X(1) \sim Poi(\lambda)$.

b. Let $T_1, T_2, \cdots$ be i.i.d. with distribution $Exp(\lambda)$. Write $S_0 = 0$ and $S_n = T_1 + \cdots + T_n$ for $n \geq 1$. Then the renewal process $X(t)$ of S_n is a Poisson process with rate λ, that is, $X(t) = \inf\{n \geq 0; S_n \geq t\}$ for $t \geq 0$.

c. The definition in the textbook in terms of $P(X(t+h) - X(t) = 1)$ and $P(X(t+h) - X(t) \geq 2)$ for small h.

• Three Poisson thinning theorems: homogeneous Poisson processes, homogeneous Poisson processes with time-dependent intensity function, non-homogeneous Poisson processes with time-dependent intensity function.

• Let $N_1(t)$ and $N_2(t)$ be independent Poisson processes with rates λ_1 and λ_2, respectively. Let $S_i(k)$ denote the time of the k-th event for $i = 1, 2$. Then

$$P(S_1(n) < S_2(m)) = P\left(X_1 + X_2 + \cdots + X_{m+n+1} \geq n\right),$$

where $X_1, X_2, \cdots$ are i.i.d. $Ber(p)$ and $p = P(S_1(1) < S_2(1)) = \lambda_1/(\lambda_1 + \lambda_2)$. The probability is the same as that of n Heads occurring before m Tails when flipping a coin with $P(H) = p = 1 - P(T)$, repeatedly.

• *For a non-homogeneous Poisson processes $N(t)$ with intensity function $\lambda(t)$,*

a. $N(t)$ has independent increments;

b. $N(t+s) - N(s)$ is a poisson random variable with parameter $m_s(t) = \int_s^{t+s} \lambda(u)\, du$;

c. Given $N(t) = n$, the n arrival times have the same distribution as that of the order

statistics of n i.i.d. random variables with pdf

$$f(x) = \frac{\lambda(x)}{\int_0^t \lambda(s)\, ds}, \quad 0 < s < t.$$

• Compound Poisson process $X(t) = \sum_{i=1}^{N(t)} Y_i$, where $N(t)$ is a Poisson process and Y_i's are i.i.d. r.v.'s independent of $N(t)$. $X(t)$ has a similar flavor of branching processes.

5. Continuous time Markov chains.

• *Matrix Q and the Kolmogorov backward equation*

The matrix Q is the derivative of the transition matrix $P(t) = (P_{ij}(t))$ at $t = 0$, that is, $Q = P'(0)$. The derivative $P'(t)$ for any $t \geq 0$ is given by the Kolmogorov backward and forward equations $P'(t) = QP(t) = P(t)Q$. Since $P(0) = I$, the the Kolmogorov backward equation implies that $Q = P'(0)$.

• Some concepts: recurrence, transience, positive/null recurrence, stationary distributions.

• A continuous time MC is irreducible iff its embedded chain is irreducible.

• A state i is recurrent/transient iff it is so in the embedded chain.

• In general, positive recurrence is not equivalent in continuous time MC and the embedded chain. However, if the state space is finite, and if i is positive recurrent in the embedded chain, it is so in the continuous time MC.

• *Limiting probabilities for birth-death processes*

Assume $S = \{0, 1, \cdots\}$ and $P_{0,1} = 1$. At state 0, by waiting exponential time $Exp(\lambda_0)$ the chain moves to 1. At state $i \geq 1$, there are two competing clocks: $Exp(\lambda_i)$ for state $i + 1$, and $Exp(\mu_k)$ for state $i - 1$. The shortest time decides the next move. Thus,

$$P_{i,i+1} = \frac{\lambda_i}{\lambda_i + \mu_i} \quad \text{and} \quad P_{i,i-1} = \frac{\mu_i}{\lambda_i + \mu_i}, \quad i \geq 1.$$

Let $P_0, P_1, \cdots$ be the stationary distribution. Then necessarily

$$\sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} < \infty$$

and

$$P_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right)^{-1}.$$

• *Limiting probabilities for $M/M/s$ as special birth-death processes*

There are three cases: $M/M/1, M/M/s, M/M/\infty$.